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ReneuBio Inc is a platform biotech company built around **AlphaForge**, **a clinical risk prediction and molecular redesign engine**. Our core premise is that many drug failures are not random; key liabilities are often embedded in the molecule, but only discovered after years and millions of dollars in development. AlphaForge surfaces those risks earlier, explains what is driving them, and helps teams decide which programs to advance, redesign, or stop. It also enables redesign around those key clinical liabilities, allowing us and our partners to generate better molecules.

Platform Advantage - Predict, Explain, Redesign: AlphaForge evaluates molecules directly from their structure in the context of their biology and disease, identifies likely liabilities, and estimates the clinical risks most likely to matter during development. In practice, it answers three questions: **Is this molecule likely to fail? Why? And what should we do next?** A key differentiator is that AlphaForge does not just generate a score; it provides **structural rationale**. This allows chemists and partners to understand the drivers of risk and make actionable development decisions, rather than relying on black-box predictions. Just as importantly, AlphaForge can guide targeted redesign around those liabilities, turning risk insight into better molecules.

Proof of Value - From Internal Program to Generalizable Capability: AlphaForge was first built to solve a problem inside ReneuBio's own CNS pipeline. In one representative program, the platform evaluated an internal lead molecule, RNB001, and discovered liabilities tied to its toxicity risk and short duration. It then guided the identification of an entirely new molecule, RNB027. The new molecule showed a ~3x improvement in pharmacodynamics in aged animals, and the design cycle was completed in under two months with minimal experimental cost. This demonstrated that AlphaForge can do more than evaluate molecules; it can help generate better ones and support smarter development decisions. We are now applying this approach more broadly, including through collaboration with **UC Berkeley's Molecular Therapeutics Initiative**.

ReneuBio Highlights

- Co-founders have authored scientific work published in ***Nature*** & ***Science***
- Proprietary **clinical risk prediction and molecular redesign engine**, **AlphaForge**
- Risk score shown to **track real-world clinical outcomes**
- Model frozen in 2025; forward predictions (2025 & 2026) continue to match observed clinical outcomes.
- Internal lead redesigned into RNB027 in under two months, with ~3x in vivo PD improvement
- Collaboration initiated with **UC Berkeley's Molecular Therapeutics Initiative**

Biology Foundation - Aging-Relevant CNS Focus: ReneuBio's advantage is not only in molecular design, but in where we start. Our founders developed screening systems in aged cells and directly in aged brains, identifying **301 targets** relevant to the aging brain, with multiple validated targets and patents filed or secured. This work, grounded in publications in **Nature** and **Science** by the co-founders, provides a strong biological foundation and differentiates ReneuBio from purely AI-driven approaches that lack age-relevant CNS biology.

Collaboration Model - Platform First, Asset Optionality: ReneuBio is not a software company. AlphaForge is deployed through collaborative engagements with biotech and pharma partners to assess and de-risk existing compounds, identify structural liabilities, support earlier and more informed portfolio and clinical development decisions, and guide redesign toward lower-risk molecules where chemistry is limiting a program. In parallel, we selectively build or option internal assets where the platform provides a clear advantage. This creates a model with near-term partner value and long-term asset upside, while remaining capital-efficient and focused.

Team and Institutional Roots: ReneuBio is led by a multidisciplinary team spanning AI drug design, medicinal chemistry, translational neuroscience, and aging biology. The team brings experience from **Stanford, Pfizer, PTC Therapeutics,** and **AbCellera**, and has advanced small-molecule programs from discovery into the clinic and to market. Our scientific foundation includes founder's work at **Stanford University**, and ReneuBio has also been part of **Berkeley SkyDeck**, reflecting both strong scientific roots and a focus on building a scalable company.